

Audience: Head of Risk Management, Lending Club

1. Problem and Stakes

Lending Club already assigns each loan applicant a letter grade based on its internal model. In this project, I build a second-opinion model that uses only application time information to predict whether an applicant would be classified as prime A-C or higher risk D-G. The target variable in the notebook is `risky_grade`, where 0 represents prime borrowers and 1 represents higher risk borrowers. This matters because an accurate 2nd opinion model could help lending club identify applicants who deserve closer review before funding decisions are finalized. The dataset is imbalanced with about 18% of applicants failing into the risky category, so model evaluation had to account for the business reality.

2. Recommendation

I recommended Gradient Boosting as the production model for Lending Club's 2nd opinion classification system. In my notebook, Gradient Boosting achieved the strongest overall ranking performance on the held out test set, with a ROC-AUC of 0.7911, outperforming Logistic regression (0.7662), the Neural Network (0.7752), and the Decision Tree (0.7137). At the default threshold, Gradient Boosting also delivered 0.8255 accuracy, 0.5778 precision, 0.2108 recall, and 0.3089 F1 on the test set. I selected the best Gradient Boosting version using validation ROC-AUC, and the best setting was `learning_rate = 0.1` and `max_depth = 3`.

The main reason to deploy Gradient Boosting is that it ranked risky borrowers better than the other three models. For this business problem, ranking quality matters because Lending Club can use the score to prioritize applicants for the review rather than relying only on a fixed yes or no decision rule.

3. Top 5 Drivers

The 5 most predictive features in the recommended Gradient Boosting model were `term`, `total_debt_limit`, `debt_to_income`, `loan_amount`, and `verified_income_Verified`.

1. Term

Longer loan terms, especially 60-month loans appeared more often among higher risk borrowers than prime borrowers. This suggest repayment horizon is an important signal in how Lending Club distinguishes stronger applicants from riskier ones.

2. Total Debit Limit

Lower total debt limit tended to appear among riskier borrowers, suggesting thinner overall available credit capacity. In business terms, applicants with less overall room in their credit profile may appear financially tiger and therefore riskier.

3. Debt to Income Ratio

Higher debt to income ratios were important in separating riskier loans from prime loans. This is consistent with the idea that borrowers already carrying heavier debt obligations may be less able to absorb another loan safely.

4. Loan Amount

Larger requested loan amounts carried a useful signal in the risky versus prime classification. A larger requested loan can increase repayment pressure and may raise concern when combined with other weaker application characteristics.

5. Verified Income Status

Verified_income_verified appeared in the models top 5 features, which suggest that the level of income documentation is related to how Lending Club distinguishes prime from riskier applicants. In operational terms, income verification appears to carry useful information beyond the raw income amount alone.

4. Tradeoff and Threshold

Although Gradient Boosting was the strongest model overall, the default .50 threshold did not catch many risky borrowers. At that threshold, recall was only 0.2108 meaning that the model identified only a small share of all truly risky applicants. To address that, I selected the operating threshold using the validation set, not the test set, by maximizing validation F1. The chosen threshold was approximately 0.18, with the best validation result around 0.185, where validation precision was about 0.3597, validation recall was about 0.7230, and validation f1 was about 0.4804.

Using that lower threshold on the test set changed the tradeoff substantially. At the chosen threshold, Gradient Boosting achieved 0.7255 accuracy, 0.3709 precision, 0.6946 recall, and 0.4835 F1, while ROC-AUC remained 0.7911. Basically lowering the threshold caused the model to catch far more risky borrowers, but it also increased the number of prime borrowers who would be flagged for extra review. For a credit risk use case, that is a reasonable tradeoff when the business views missed risky loans as more costly than reviewing extra applicants.

5. Operational Recommendation

Lending Club should use the Gradient Boosting model as a triage tool, not as an automatic rejection engine. Applicants about the chosen risk threshold should be sent to enhanced review, such as added documentation checks, pricing review, or manual underwriter attention. This would help the company focus resources on the applicants most likely to be higher risk while still preserving human oversight.

A key caveat is that the model is based on historical application patterns and the current cleaned dataset. If a borrower's behavior, underwriting standards, or documentation practices change, performance may drift over time. Lending Club should monitor results after deployment and periodically retrain the model or adjust/ recalibrate the threshold as conditions change.